Grammaticality de-idealized

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Abstract

Why is *I have 25 years* easy to understand but ungrammatical in English, while *Colorless green ideas sleep furiously* is grammatical despite its odd meaning? I argue that grammaticality research often runs together four questions. Does the language provide an analysis for the expression? Do the grammatical contributions of its parts fit together? Does it mark the contrasts required in this context? Is the construction established in this speech community and situation? Processing difficulty and implausibility are separate: they can lower acceptability without changing grammatical status. The Operator-Value Model of Grammaticality (OVMG) develops these distinctions and models how use, competition among alternatives, and forgetting change the constructions available to a population; a separate supplement gives the mathematical specification. The model predicts that rarely encountered constructions should respond more to exposure and framing than forms consistently displaced by an established competitor; that the absence of a form is evidence of ungrammaticality only when speakers repeatedly chose an established alternative in contexts where the missing form would have been a plausible choice; and that, as a contrast becomes moribund, the evidence sustaining it should weaken before its categorical status changes. Simulations show that sharply separated population patterns can arise under specified adoption conditions, but not from making an analyst's evidence-based estimate more precise alone. The proposal replaces the ideal speaker's single grammaticality fact with distinct, testable claims about structure, convention, speaker experience, and change.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization; usage-based grammar

AI use. The large language models Claude, ChatGPT/Codex, Gemini, DeepSeek, and Ox Alpha served as drafting and editing aids throughout the preparation of this paper. I remain responsible for all theoretical claims, arguments, errors, and interpretive choices.

Aristotle provided formalization assistance. Model outputs used in adversarial review or formalization were checked, where applicable, by independent recomputation, local proof replay, or executable tests.

Introduction

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

(1)
a. **Can the have running?*
b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
c. **I've finished it yesterday.*
d. *?I saw Joan, a friend of whose was visiting.*
e. *The bread the baker the apprentice helped made is delicious.*
f. A: *How old are you?* B: **I have 25 years.*
g. **Which did you buy car?*

While all might receive asterisks in many analyses, they represent fundamentally different types of unacceptability. Some constructions, like (1a), fail to establish any form–value relationship in English. Others, like (1c), involve incompatible operator contributions: the present perfect's reference-interval contribution doesn't unify with *yesterday*'s deictic anchoring. Still others, such as (1d), show gradient or indeterminate status. (1e) illustrates what might be called predicted grammatical but reliably rejected: constructions that linguistic theory predicts should be grammatical but that are consistently rejected by native speakers in ordinary judgment tasks. Still others, like (1f), are readily interpretable and grammatical in related languages but lack licensing in English age predication. And a few, like the left-branch extraction of (1g), seem to be ruled out entirely, despite being apparently short and interpretable.

Different traditions emphasize different pieces: formal approaches focus on abstract principles, usage-based theories on frequency and entrenchment, processing accounts on cognitive constraints. Each captures part of the picture, but none unifies categorical constraints, gradient acceptability, cross-linguistic variation, and change over time. I'll argue that the obstacle is a shared idealization, and that removing it reorders the questions.

The de-idealization in the title is specific. Early generative grammar idealized grammaticality to the knowledge of an ideal speaker–listener in a completely homogeneous speech community (Chomsky, 1965, p. 3), treating variation, uncertainty, and change as noise around a categorical competence fact. That idealization bought tractability, but it priced out the phenomena in (1). Following Goodman (1955), removing it reorders the questions: ask first what the category GRAMMATICAL lets us predict (its projectibility, what observing some features licenses us to expect about others), and only then what worldly profile supports those predictions and what stabilizes it (Boyd, 1999; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). Mechanisms matter because they explain why the projection holds, not because naming them settles the kind.

This paper develops that answer as the Operator-Value Model of Grammaticality (OVMG). Some grammatical contrasts tell interlocutors how to treat an utterance: as a question rather than an assertion, as committing one participant rather than another, or as linking one referent to another across clauses. I call these contrasts OPERATOR values and their contribution a PUBLIC-UPDATE ROLE. They form closed paradigms of instructions rather than a class defined by expressive substance, so tone, particles, word order, and inflection all count when they realize such a contrast

(Reynolds, 2026c). The framework rests on three premises, which track the order just named: profile, projection, and support.

1. Profile. A grammaticality claim is shorthand for a profile: a bundle of answers to several separable questions about a candidate in a situation. Is there any available analysis for it? Do its grammatically encoded contrasts agree with one another? Are the contrasts required in that situation supplied? Is the pattern a conventional resource for the relevant community, dialect, or register? And, separately, does it merely feel bad because it's implausible or hard to process? Section 3 develops these as coverage, compatibility, saturation, licensing, and subjective read-out. The examples in (1) don't instantiate one kind of "ungrammaticality"; they occupy different profiles.
2. Projection. The profile earns its standing by what it lets us predict, not by the mechanisms that hold it together: which rejections should soften under repeated exposure and which resist (1d) vs. (1g), how faithfully forms transmit, and how they change over time.
3. Support, tested separately. The dynamics module specifies candidate sufficient conditions under which usage could stabilize a population licensing state; conversational repair is a candidate *controller*, testable by its intervention signature (§4.4); and the subjective read-out of ungrammaticality is separate from grammatical status: it comprises gradient anomaly and confidence signals.

Together, the premises explain why the traditions have talked past one another. The profile premise preserves the formal insight that constraints can be systematic while letting categorical and gradient cases share one architecture, in line with Francis's (2022) case for building gradience into core grammatical theory. The projection premise keeps the account testable: low-evidence constructions should respond to exposure and framing differently from high-opportunity preempted forms. The support premise explains why formal licensing, processing cost, and subjective read-out can move together without being the same thing.

The argument proceeds in three moves. Section 1 shows why no single scale of acceptability can distinguish structural failure, incompatible grammatical instructions, community convention, and processing difficulty. Sections 2–4 then build the alternative: first a reader-facing profile for diagnosing a token, then a separate account of how competition, learning, and repair may change a population's repertoire. The remaining sections ask what that architecture buys. Section 5 sets its predictions against generative, constructional, usage-based, logicality, and relevance-theoretic accounts; the final sections state the empirical limits and the programme needed to decide among them. The accompanying formal supplement contains the notation, derivations, simulations, and machine-checked structural results.

1 The impasse in grammaticality theory

The concept of grammaticality remains elusive despite its centrality to linguistic theory. After decades of research, the field still lacks a unified account of what makes an utterance grammatical or ungrammatical. The impasse persists because grammaticality is often treated as a single phenomenon, when judgments actually reflect several questions at once: whether a covering assembly exists, whether its operator contributions are compatible, whether the situation's obligatory dimensions are saturated, whether the relevant constructions are licensed, and how the resulting token feels to a speaker.

1.1 The problem of structural viability

The first challenge in defining grammaticality is distinguishing structures that admit no viable analysis from those that fail for other reasons. Chomsky (1957), building on formal production systems developed by Post (1943), treated grammaticality as a categorical property defined by membership in a set of well-formed strings. That categorical view captures the intuition that some strings, like (1a), resist structural analysis entirely:

(2) **Can the have running?*

Here, the input crashes the structural analysis independent of meaning. The "categorical" view correctly identifies that a viable structural analysis is necessary for grammaticality. But raising this single condition to the *definition* of grammaticality leaves formal approaches struggling with evidence that speakers consistently provide gradient judgments for structures that are clearly analyzable but degraded. The competence-performance distinction (Chomsky, 1965) attempted to preserve categorical grammar by attributing gradience to processing limitations. But as Schütze (2016, p. 71) notes, this often leads to circularity, where problematic results are dismissed as performance artifacts. The OVMG framework resolves this by recognizing that while structural coverage–the existence of some assembly covering the relevant form–value pair–is a necessary, categorical prerequisite, it isn't the sole determinant of grammaticality.

1.2 The problem of operator compatibility

The second tension arises from the interaction of form and value. Chomsky's famous example (1b) shows that operator-valued form–value relations can be intact even when lexical plausibility fails: the sentence is structurally a declarative, the subject is correctly identified both positionally and through agreement, and the tense suggests a general claim. What breaks is the plausibility of the construal, not grammatical status. Conversely, (1c) **I've finished it yesterday* shows that an intended meaning can be readily recovered even when an operator-valued form–value relation fails. Here, the clash is between the present perfect's temporal contribution and the deictic anchoring contributed by *yesterday*.

Generative semanticists like Lakoff (1971) and McCawley (1968) demonstrated early on that many constraints have semantic motivations. Construction Grammar (Goldberg, 1995) has further shown how constructions carry meanings that interact with lexical semantics. Novel uses that align with established constructional meanings (like *She texted him the address*) are accepted, while those that clash with a construction's licensed value (like **She disappeared him the evidence*) are rejected. A complete theory needs a hard compatibility condition for conflicts among grammatically encoded operator contributions, while keeping lexical implausibility and processing cost in the subjective cost vector rather than turning every odd construal into a status failure.

1.3 The problem of situational licensing

Finally, even structurally viable forms with readily recoverable interpretations may be ungrammatical if they violate community conventions. Sociolinguistic research (Labov, 1972) demonstrates that grammaticality references community norms. Cross-linguistic variation shows that each language community conventionalizes particular form–value mappings through historical processes. These conventions become entrenched through statistical preemption (Goldberg, 2011): speakers learn that certain forms are ungrammatical precisely because they encounter alternative expressions in contexts where the ungrammatical form would otherwise be expected.

Consider (1f), repeated here:

(3) A: *How old are you?* B: **I have 25 years.*

This utterance is structurally viable and easy to interpret, but it's categorically rejected in English. The same form is grammatical in French (*J'ai 25 ans*) and Spanish (*Tengo 25 años*); the constraint is community-specific. Usage-based approaches (Bybee, 2006) emphasize that such rejection arises from the absence of situational licensing. The English-speaking community has conventionalized a different way to express age, so the *have*+age frame is preempted by the BE-frame in a dense niche rather than merely being unattested.

These problems don't reduce to one scale. Structural coverage, operator compatibility, saturation, licensing, and subjective read-out have different diagnostics and different evidence streams. Acknowledging them as distinct but interacting routes makes it possible to move beyond the impasse of trying to force all grammaticality judgments into a single "competence" definition or a purely probabilistic usage model.

2 The Operator-Value Model of Grammaticality

The term VALUE is used here in the Saussurean sense of *valeur* (Saussure, 1916): the identity of a linguistic unit is constituted not by intrinsic properties but by its position in a system of contrasts, what it patterns with, what it opposes, and what interpretations it makes available. This isn't feature-value assignment or decision-theoretic value. Throughout, I treat grammatical knowledge as a conditioned form–value relation; when this relation is sufficiently stable in a communicative situation, with a dominant value reliably recoverable and socially licensed, I refer to it informally as an established form–value relationship.

The model's name marks its unit: operator values (closed-paradigm, update-configuring contrasts). A companion operator-stratum paper develops this boundary in full and argues that it follows public-update role rather than expressive substance: tone, particles, word order, inflection, and suprasegmental material all fall within its scope when they realize an operator contrast (Reynolds, 2026c).

Here a self-contained working test suffices, because the concept has to stand on its own inside this paper: it fixes the scope of hard compatibility, the operator-specific repair prediction, and the appendix on Turkish suffix harmony (§A).

A conventionalized contrast is an operator contrast when it meets two conditions jointly: it comes from a *closed* paradigm, and the paradigm is *configured to public update*. A public update is a change in what participants treat as being on the conversational record: whether something is asserted or questioned, who did what to whom, when it holds, what follows from it, and which referents or commitments later contributions can pick up. Clause type, polarity, tense and aspect anchoring, agreement and role allocation, scope, and reference tracking are illustrative rather than exhaustive cases. The condition applies across a paradigm's occasions of use, not to the information carried by every token. Mis-setting a value can change the update instruction even when the intended message is recoverable; selecting an excluded exponent can instead leave the intended value intact. Comprehension probes can test the distinction directly by asking whether changing the contrast changes hearers' answers about the public record (Reynolds, 2026c).

Opportunity density modulates the resulting licensing profile: contrasts that recur often provide more evidence and can become more categorical and resistant to exposure. But opportunity doesn't determine operator membership. A moribund case distinction can remain an operator while its licensing profile destabilizes. A frequent collocation or a head-specific complement construction can

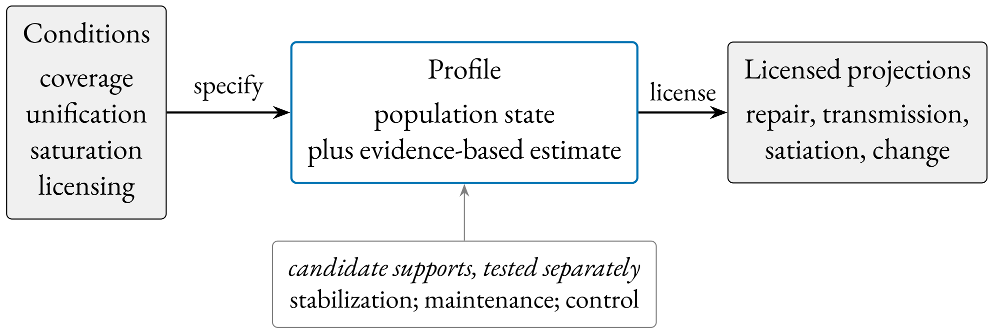

Figure 1: The projectibility-first order of explanation. The constitutive conditions specify a licensed-assembly population profile. Evidence about that profile, including how settled the evidence is and how much speakers differ, licenses the projections on the right. Stabilization, maintenance, and control are separate world-side hypotheses (below), each requiring its own evidence.

be categorically licensed or excluded without becoming an operator. Closure and public-update role define the operator stratum, not grammaticality as a whole. The narrower falsifier is response-based: after closure, opportunity, and competitor strength are matched, a contrast that doesn't configure the public update shouldn't produce the same update-confusion and update-oriented repair profile as one that does. If it does, public-update role adds no explanatory boundary beyond constructional licensing and entrenchment.

Four token-level questions remain separate: which construction does the community license; does its paradigm have a public-update role; did the token select the intended grammatical value; and did it use a licensed exponent?

English **depend of* fails the first question through a head-specific complement convention without mis-setting an operator. **Goed* and the Turkish harmony cases in Appendix A select the intended operator value but fail the last question. These dimensions can intersect; they aren't exclusive error classes.

The architecture rests on four constitutive conditions: structural viability, operator-value compatibility, saturation of obligatory dimensions, and situational licensing. Together they specify a PROFILE: observing part of that pattern should license further expectations about repair, learning, transmission, exposure, and change (Reynolds, 2026a). Figure 1 lays out the order of explanation. The conditions specify the profile; the profile supports testable projections; candidate historical and corrective processes are investigated separately as possible supports for it.

The model also needs names for recurrent INSTABILITY MODES: ways of defeating coverage, compatibility, saturation, or licensing, with processing and prescriptive factors modulating how grammatical something *feels* without directly fixing grammatical status. Section 3 develops the four questions in ordinary language; the supplement supplies their exact representation.

2.1 Diagnostic categories

The categories below are labels for common routes to ungrammaticality, not primitives of the model. Each traces to one or more constitutive conditions, as the reader-facing account in §3 makes explicit.
FORM–VALUE RELATIONS. Grammaticality depends on stable links between forms and the operator values they realize within communicative situations. That form–value relations exist at every level of

grammatical organization, including morphemes and abstract syntactic patterns, is a core insight of Construction Grammar (Goldberg, 1995). OVMG identifies an operator stratum within grammaticality: closed-paradigm, update-configuring contrasts whose expressive resources help determine how an utterance enters public coordination. Opportunity density affects how concentrated their licensing profiles become.

Operator exponents, like words, are typically polysemous: a single form participates in multiple form–value relationships, usually with one dominant value (Kilgarriff, 2004). The English past tense, for instance, usually denotes past time, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow...*). What matters for grammaticality is whether a stable relationship exists within the relevant communicative situation, not whether the form has a single fixed meaning.

Neuroimaging work provides converging evidence that form and meaning aren't processed by wholly sealed modules. Fedorenko and colleagues distinguish language-selective regions from neighbouring domain-general systems while showing that syntactic and semantic manipulations both recruit the language-processing architecture (Fedorenko et al., 2011, 2012, 2024). The processing claim here is modest: the subjective read-out can reflect interacting processing demands even when grammatical status remains intact.

Following Wiese (2023, p. 3), grammaticality is conditioned by regularities within a "communicative situation". The emergence claim defended below is narrower: categoricality arises through the licensing dynamics in §4.3, not through an unanalyzed appeal to usage.

Communicative situations aren't static or mutually exclusive. A speaker orients to overlapping contexts that shift on multiple dimensions: the immediate situation (formal meeting vs. casual chat), durable social positions (regional, generational), and the norms they're orienting to (which can shift with audience). This fluidity explains style-shifting, dialect formation, and the gradual acceptance of initially marginal constructions.

Early child language shows this. Toddlers produce utterances that deviate from adult norms but remain internally consistent within the child's developing system. A toddler in a monolingual English household might say:

(4) *Ava cookie.* (intended as 'Ava = I want a cookie')

Although this differs from adult English norms, it may not be perceived as ungrammatical by the child's regular caregivers. Used with the same pragmasemantic force among anglophone adults, the construction would be judged ungrammatical.

Multiple modal constructions provide another clear example of variety-relative grammaticality:

(5) *I might could help you with that.*

This combination of modal auxiliaries is systematically possible for some American English speakers, who can productively generate similar constructions (Morin & Grieve, 2024). Speakers from communities where only single modals are grammatical, though, typically reject such combinations as ungrammatical.

A similar pattern appears in code-mixing among bilingual speakers, where combinations of forms from different languages can be grammatical within that bilingual community's norms but not without. Consider a Spanish-English bilingual speaker who uses a Spanish progressive auxiliary with an English lexical verb:

(6) *Ayer, estábamos lifting en el gym durante una hora.*
yesterday be.IMPF-1PL lifting in the gym for an hour
'Yesterday, we were lifting in the gym for an hour.'

Within the right communicative situation, this utterance is grammatical. The Spanish auxiliary *estábamos* combines with an English participial form *lifting* to form the progressive aspect. This cross-linguistic combination of morphology and a lexical verb is consistent with local norms, where code-mixed utterances are common and meaningful. In contrast, in a standard monolingual Spanish setting, where fully Spanish progressive structures are licensed (*estábamos levantando pesas*), speakers may judge the example in (6) as ungrammatical. The use of intransitive *lifting*, specific to the gym community, further illustrates just how localized grammaticality judgments can be.

This doesn't mean bilingual communities simply accept any combination of languages. As Toribio (2001) reports, Spanish–English bilingual speakers judge examples like (7) as unacceptable, showing that even in bilingual communities, there are systematic constraints on which language combinations are permitted.

(7) * *Los enanitos intentaron pero no succeeded in awakening Snow White* (Toribio, 2001)
the dwarfs try.PST-3PL but not succeeded in awakening Snow White
'The dwarfs tried but didn't succeed in awakening Snow White.'

In a slightly different case, as a second-language speaker of Japanese, I used to say

(8) * *Kawaii da.*
cute COP.NPST
'(That)'s cute.' (intended)

Conventionally, though, plain *i*-adjective predicates in Japanese don't take the plain copula *da* in this frame. The issue is exponent selection: the predicate's value is recoverable, but the community doesn't license that copular exponent there. The example shows why acquisition history matters: shared routines stabilize form–value relations for a community, while individual trajectories can still pull judgments away from that norm.

The specific linguistic context can matter too:

> To take an obvious case which Jerry Morgan [(1973)] discussed recently, certain combinations of words are extremely strange if presented in isolation but are perfectly normal as answers to certain questions. *Spiro conjectures Ex-Lax* would generally be felt to be unintelligible if presented out of context but is a perfectly normal answer to the question *Does anyone know what Mrs. Nixon frosts her cakes with?* (McCawley, 1974, 252, italics added)[1]

These examples show why grammaticality has to be indexed to communicative situation. Conversants who differ in immediate situation, social position, or the norms they orient to may disagree on whether a form is grammatical, and both may be right within their respective contexts. The linguist's task is to determine whether a form is grammatical from some position in that space, or systematically excluded from all of it.

[1] The humour derives from political tensions of the Watergate era, when both Vice-President Spiro Agnew and President Nixon would ultimately resign from office.

The stability of form–value relations within speech communities can be empirically modeled, as demonstrated by Blythe and Croft (2009). Their utterance selection model treats speech communities as networks where speakers track and reproduce linguistic variants based on their interactions. When applied to dialect formation, these models show how competing linguistic variants spread and eventually stabilize, with initial variant frequency strongly predicting which form will prevail.

In different languages, different distinctions become conventionalized as grammatically obligatory. These obligatory contrasts reflect historically stabilized patterns in which meaning distinctions are systematically marked. Over time, usage establishes grammatical constructions that reliably signal these distinctions. As a result, what counts as a grammatical necessity in one language may be optional or absent in another.

The progressive aspect provides a useful example. In Standard English, it isn't optional but an obligatory grammatical marker for ongoing, incomplete actions:

(9) *She is studying right now.*

Here, the progressive form *is studying* isn't just a stylistic choice. It's the recognized, grammatical way to convey the appropriate aspectual meaning in this frame.

In French, by contrast, the progressive aspect isn't a grammatically mandatory distinction. Speakers can signal ongoing activity through adverbs or periphrastic constructions, but standard French doesn't have a dedicated progressive form. The sentence:

(10) *Elle étudie maintenant.*
'she studies/is studying now'

can comfortably describe a currently ongoing action without any need for special morphology. In Standard French, this aspectual distinction isn't conventionalized as an obligatory operator contrast. English grammar treats the contrast as obligatory; French doesn't. On this account that difference isn't a primitive of either grammar: §4.5 models obligatoriness as the limit of preemption applied to zero-marking, so the English–French contrast is a prediction of the dynamics, not a stipulation they inherit.

The same holds for evidentiality, the grammatical marking of information sources. In Turkish, evidentiality is systematically realized through verb forms and particles that distinguish between directly witnessed events and those inferred or reported:

(11) *Gelmiş*
'He/she came (apparently)'
(meaning the speaker wasn't a witness but inferred or heard about it.)

In Turkish, evidential distinctions are conventionalized as obligatory operator contrasts. A speaker can't simply omit evidential marking without sounding ungrammatical.

English speakers can say *I heard that he arrived* or *He must have arrived*, but these are optional lexical or modal resources rather than required elements of the grammar. In English, evidential distinctions aren't conventionalized as obligatory operator contrasts. The same mechanism covers the Turkish case on a different dimension (§4.5).

Kilani-Schoch and Dressler (2005) demonstrate this principle systematically in their analysis of verbal morphology across French and other Romance languages. They show how apparently similar verbal systems can grammaticalize quite different semantic distinctions as obligatory. The contrast reflects different conventions about which meaning distinctions are systematically marked. For instance, while both French and Italian mark aspect morphologically, they differ in which aspectual distinctions are grammaticalized versus left to optional lexical expression.

Establishing a form–value mapping is necessary for grammaticality, but it isn't sufficient. Speakers routinely interpret strings that still count as ill-formed in the relevant communicative situation. That flexibility makes licensing, not interpretability in principle, the central question.

The conditions do different work. Coverage asks whether any assembly can be built; compatibility asks whether its operator contributions unify; saturation asks whether the obligatory dimensions for the situation are set; licensing asks whether the population treats the relevant constructions as repertoire. Processing costs, recovery costs, and lexical implausibility can still make a licensed assembly feel bad, but they enter selection and the subjective cost vector rather than replacing status. The interaction predicts why **Furiously sleep ideas green colorless* elicits universal rejection while (1e) feels bad in everyday quotation but is accepted as "grammatical but hard to process" once speakers are walked through the intended structure.

Nor does convergent usage by itself settle the matter. Defining grammaticality as "whatever speakers accept" would be circular: it couldn't flag contested innovations, explain stable dialectal differences despite mutual intelligibility, or distinguish licensed forms from performance errors that pass unnoticed. Separating coverage, compatibility, saturation, licensing, and subjective read-out avoids trivializing grammaticality while still treating durable population-level licensing as decisive for which relationships endure.

Operator-value compatibility. Beyond the existence of form–value relations, grammaticality requires compatibility among the operator values those relations realize. This condition captures cases where individual elements are well formed but their combination creates conflicts among grammaticalized contrasts activated in the communicative situation.

Consider the temporal incompatibility in:

(12) * *I've finished it yesterday.*

Here, the present perfect construction contributes a reference interval anchored to present relevance, while *yesterday* contributes a deictically past interval. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows failed unification between grammatically encoded temporal contributions.

The age example belongs elsewhere:

(13) * *I have 25 years.* intended as 'I'm 25 years old'

In English, *have+years* is a licensed frame for duration, remaining time, or experience, but ordinary age predication is licensed through a different frame. The intended value is readily recoverable, so the failure isn't an operator-compatibility failure. It's a licensing failure for an age-predication construction in English, even though cognate-looking frames are licensed in French and Spanish.

Sometimes compatibility failures involve conflicting information-packaging requirements:

(14) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

The clash is in grammatically encoded information packaging: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit compatibility. The compatibility condition treats operator-value conflict as failed unification among partial operator assignments: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-packaging instructions where a construction grammatically encodes focus or backgrounding, and indexical contrasts only when they've entered a closed accountable repertoire. This makes compatibility partly derivative of licensing facts at the feature level, keeping arbitrary lexical implausibility out of grammatical incompatibility.

Processing constraints. Language processing engages both dedicated language networks and domain-general cognitive systems (Fedorenko et al., 2024). When constructions overload these systems, particularly through multiple long-distance dependencies or heavy embedding, they may trigger feelings of ungrammaticality despite being structurally well-formed.

(15) *The bread the baker the apprentice helped made is delicious.*

Evidence for these constraints comes from multiple sources. Studies of sentence processing show that dependencies spanning multiple intervening elements increase processing difficulty, with new referents between dependent elements compounding memory load (Gibson, 2000, 2026). In (15), while each individual relation (like *the apprentice helped* and *the baker made*) is interpretable in isolation, their nested combination overwhelms incremental processing. The construction is grammatical, but excessive bridging costs from its nested structure trigger feelings of ungrammaticality.

Rather than reflecting simple memory limitations, processing constraints emerge from the interaction between specialized language networks and other cognitive systems (Fedorenko et al., 2024). On this processing-level interpretation, what speakers experience as difficulty may reflect demands distributed across specialized brain systems rather than a single cognitive bottleneck.

Dependency locality is the best-documented such cost. Integration difficulty rises with dependency length and peaks at loci of long-distance integration, with locality accounting for substantial region-level variance in reading times in Grodner and Gibson (2005). Because the effect recurs across processing studies, the supplement represents dependency locality separately rather than hiding it inside a residual term. High-locality constructions feel worse even when structurally well-formed, and because they're harder to produce and parse, speakers tend to avoid them.

That avoidance is a pathway from processing to change, developed in §2.5 and §4: lower-processing forms are easier to store and reuse, so simpler patterns can spread and conventionalize. The processing evidence supports this as a plausibility claim about one route into stability, not a direct neural explanation of grammatical change.

Processing constraints also interact with the licensing state without becoming part of status. A construction supported by thin or conflicting licensing evidence may be judged more harshly when it's also hard to process. Conversely, a highly licensed construction can remain grammatical despite considerable processing demands, even while those demands keep its anomaly signal high.

Socio-pragmatic indexicality. The value of a construction extends beyond semantic content to include socio-pragmatic dimensions: how linguistic forms index aspects of social context, speaker identity, group membership, stance, or interpersonal relationships (Eckert, 2012; Silverstein, 1976).

In many varieties of Latin American Spanish, for example (e.g. the Río de la Plata region), speakers use *vos* and its associated verb forms instead of the *tú* forms used elsewhere in the Spanish-speaking world (Bertolotti, 2016):

(16) *¿Vos querés un café?*
you.SG want.2SG-VOS a coffee
'Do you want a coffee?'

Here, the use of *vos* rather than *tú* denotes the second-person singular hearer (the person being offered a coffee) and indexes the speaker's regional identity and familiarity with the local dialect. This indexical value may convey closeness, solidarity, or membership in a particular geographic and social community.

This situational view of grammaticality can manifest asymmetrically. Speakers from the Río de la Plata region may view a conversational situation as accommodating both their own norms and those of *tú*-using interlocutors; the situation itself can encompass both *vos* and *tú* as grammatical options. But speakers from *tú*-only regions might conceptualize the same situation more restrictively, defining it in a way that categorically excludes *vos* as a grammatical possibility. This asymmetry doesn't depend on different understandings of the forms themselves, but can arise from different ways of defining what the communicative situation allows, influenced by the indexical values attached to *vos* versus *tú* in their respective communities.

Phonology can carry constructional indexical value, but it bears on grammaticality only when it realizes an operator contrast or enters an operator-exponent licensing condition. Babel (2025) provides an example. In a study conducted in Bolivia, participants were presented with audio stimuli where only the vowels were manipulated to reflect either a highland or lowland accent. This presented participants with incongruent identity cues (e.g. vowels from one accent along with consonants from the other). But this didn't trigger feelings of ungrammaticality. Instead, vowel contrasts activated expectations about consonant features and discourse markers, and some participants "hallucinated" identity-linked features that weren't present in the signal.

So a construction's value reaches beyond semantic features to socio-pragmatic aspects that shape how speakers and hearers negotiate authority, identity, solidarity, and other interpersonal relations. Recognizing these indexical dimensions helps explain why certain forms feel natural and grammatical to some speakers and out of place or even ungrammatical to others.

SITUATIONAL LICENSING. Even well-formed constructions with clear meanings may be judged ungrammatical if they lack conventionalized licensing. This condition concerns whether a form–value relation is licensed within the relevant communicative situations.

(17) * *We sheared three sheeps.*

The regular plural marking has a compositionally recoverable meaning and is structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational licensing operates independently from the other conditions. A construction may have complete operator compatibility and minimal processing demands but still be excluded because a different form has been conventionalized for that meaning. Extreme rarity makes the distinction clear. Some constructions are so infrequent that speakers lack a shared consensus about their status.

The independent relative genitive pronoun *whose* provides an example, being so unusual that Hankamer and Postal (1973) deem it non-existent. The contexts that license this construction require the simultaneous convergence of distinct pragmatic and syntactic conditions (sufficient accessibility of both possessor and possessum, the appropriate information structure, and an environment allowing ellipsis), which are seldom met all at once (Reynolds, 2024):

(18) ? *I saw Joan, a friend of whose was visiting.*

(adapted from Huddleston & Pullum 2002, p. 472)

In the COCA check reported here, no token of the construction appears (Davies, 2024). That absence supplies little negative evidence, because the relevant opportunity count is tiny: contexts requiring an independent relative genitive are vanishingly rare. Speakers have little evidence either way: neither positive tokens nor the systematic absence that would accumulate if the construction were regularly passed over. The evidence supports high *uncertainty* rather than confident rejection. Some speakers can draw an analogy with similar constructions and accept examples like (18); others remain unsure; still others, lacking sufficient exposure, can't construct a stable form–value relation at all. Even among those who grasp the construction analytically, Shain et al. (2020)'s account predicts that its unexpectedness would trigger high surprisal, compounding the uncertainty with processing difficulty.

This contrasts with putative preempted cases like left-branch extraction, discussed next, where the relevant niche appears often and the construction is systematically untaken. The classification depends on a further quantity, though: how often speakers would choose the discontinuous form if it were stipulated to be licensed.

Apparent categorical exclusions. Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

(19) * *Which did you buy* [__ *car*]*?*

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of "ungrammaticality".

Under OVMG, such patterns are candidate preempted gaps: situational licensing is driven toward zero by persistent preemption when the communicative job is common, an established competitor wins, and the counterfactual choice probability of the missing form is high. For English left-branch extraction, the natural competitor is *Which car did you buy?*. The corpus absence is suggestive but not decisive; the missing ingredient is independent evidence about how often speakers would choose the discontinuous form if it were available under permissive framing. Additional processing penalties, such as systematic garden-path reanalysis when the bare interrogative phrase is encountered, can strengthen the subjective categoricality without any independent structural constraint. The formal dynamics appear in §4.

Cross-linguistic comparison doesn't threaten this analysis, because the dynamics are defined over a language's own niches. Left branch is a descriptive class tied to a particular phrase-structure analysis, not a comparative concept (Haspelmath, 2010); whether discontinuous nominals in case-rich, article-less languages instantiate the same configuration as the English case is far from clear, and the companion corpus study deliberately brackets the comparison (Reynolds, 2026d). The model instead predicts which opportunity structures should support similar patterns. Discontinuous nominal orders should be learnable where inflectional marking makes the dependency recoverable, a productive discontinuous-order pattern supplies scaffolding, and a discourse niche makes the separated element useful. English meets none of the three, so the candidate has no scaffolding of its own and the contiguity-preserving competitors win every opportunity.

Several diagnostic criteria help identify these preempted-gap constructions:

1. Persistent non-licensing: The construction doesn't improve under ordinary exposure or framing while the high-concentration non-licensing state remains in place.
2. Categorical rejection: Speakers find the constructions simply impossible, with high confidence rather than mere uncertainty.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn't improve acceptability.

Prediction: If LBE is a high-concentration preempted gap, framing tokens as intentional dialectal or ingroup resources should do little to move judgments; the same manipulation should matter more for low-concentration cases such as independent relative *whose* or defective cells. A substantial framing effect for LBE, or independent evidence that speakers would rarely choose the discontinuous form even if it were available, would count against the preempted-gap diagnosis.

If no covering assembly is available at all, the diagnosis isn't a further residual constraint but structural coverage failure: the existential over assemblies is empty. Treating classic cases like LBE as driven to zero under preemption keeps the constitutive core minimal and pushes the explanatory work into the dynamics of situational licensing, while reserving true coverage failure for cases with no well-typed route through the constructional inventory.

WINNERLESS CELLS: PARADIGM DEFECTIVENESS. Preempted gaps have a winning competitor. Defective paradigm cells don't. They leave a high-opportunity cell empty without any single form winning it. A clearer case is the Russian first-person singular non-past of *pobedit'* 'win': speakers avoid the expected cell and resort to periphrasis instead. The English *amn't* gap is diagnostically useful but less clear, because broader niches include *I'm not* as a competitor, while narrower contracted-negator niches make the cell look winnerless (Broadbent, 2009; Pertsova, 2016; Pertsova & Kuznetsova, 2015; Sims, 2023; Thoms et al., 2025).

In the Russian case, the niche is common and candidate forms are readily identifiable, but no candidate accumulates enough positive evidence to settle as the form. The source can be structural uncertainty, lexical restriction, learning dynamics, or avoidance, with the mixture varying across cases (Sims, 2023). In this account, defectiveness is a third licensing profile: high opportunity with evidence divided among candidate forms, leaving no candidate stably established or confidently excluded.

The phenomenology matches: speakers report not knowing how to say it (ineffability), not the confident "that's wrong" of a preempted gap. §4.5 represents this profile with an avoidance attractor plus low candidate support: avoidance absorbs the cell's opportunities, and because avoidance is nearly uninformative about any particular candidate, no candidate accumulates confident negative evidence either. The satiation-framing prediction follows: defective cells should behave like low-evidence forms, movable under framing, not like preempted gaps.

2.2 DIAGNOSING (UN)GRAMMATICALITY AT A GLANCE

For exposition, the recurrent instability modes can be read as a short decision tree (the four constitutive questions appear in §3):

Condition	Canonical outcome
no covering assembly	NONSENSE (**Can the have running*)
operator-value clash (unification failure)	compatibility failure (**I've finished it yesterday*)
obligatory dimension unset	saturation failure (unmarked progressive in English)
rarely encountered, low certainty	community-novel (*friend of whose*)
systematically preempted, high confidence	preempted gap (left-branch extraction if normed)
high opportunity, candidates split	defective cell (*pobedit'* 1SG)
high processing cost, structure recoverable	transiently ill-felt (centre embedding)
otherwise	grammatical

Lexical-semantic oddity without an operator clash (*colorless green ideas*) routes to "grammatical"; its strangeness is a construal-plausibility cost in the subjective read-out (§3.5), not a status fact.

2.3 Patterns of (un)grammaticality

These categories aren't points on one severity scale. They name different routes through the same architecture. A token can fail because no assembly covers it, because its operator contributions don't unify, because an obligatory dimension is unsaturated, or because the pivotal construction isn't licensed in the population. Separately, a licensed token can feel bad because processing, plausibility, or prescriptive dissonance pushes the anomaly channel away from zero; confidence tracks how settled the relevant status condition is.

This matters for apparent gradience. Intermediate acceptability ratings can come from unsettled licensing evidence, from a strong anomaly response to an otherwise licensed form, or from disagreement about the conditioning state. They aren't automatically evidence for soft compatibility. Recoverability of the intended interpretation can affect attribution or repair, but it can't make an operator clash grammatical. Conversely, high processing cost can make a licensed construction feel bad without weakening the evidence that the community licenses it. Section 3 separates these routes; the worked examples in Section 3.8 show how they project differently.

2.4 The subjective experience of grammaticality

The constitutive conditions determine conventional grammatical status (§3.3); speakers register putative violations through read-outs and judgments that can diverge from it.[2]

THE SUBJECTIVE READ-OUT. The familiar "feeling of ungrammaticality" is the anomaly side of a broader metacognitive read-out. It parallels other well-studied metacognitive feelings like the FEELING OF KNOWING (FOK; Hart, 1965), which arises when one feels certain of knowing something but can't retrieve it.

[2] The distinction between grammatical status and the subjective read-out has a philosophical analogue in Cavell's (1969) distinction between knowledge and acknowledgment. For Cavell, acknowledgment isn't a separate capacity from knowing but an inflection of it, bearing knowledge toward the situation that produced it. Speakers don't merely *know* that a string is ungrammatical; they *register* it as a violation, with metacognitive consequences. For LLM-based grammaticality research, the distinction blocks an easy inference: LLMs can produce judgments that correlate with human ones, but whether they acknowledge the norms they're applying or merely process patterns that correlate with those norms remains an open question (Techio, 2026).

There isn't a positive feeling of grammaticality, just as there isn't a feeling of having enough oxygen, only the negative feeling registered in its absence. The anomaly channel, then, is the negative response triggered when an utterance violates expected form–value patterns. The confidence channel tracks how settled the speaker takes the relevant status condition to be.

This notion echoes Edward Sapir's "form-feeling", an often unconscious grasp of language patterns (Sapir, 1921, 1927). Evidence from aphasia supports the same separation. Patients with Broca's aphasia, who produce agrammatic speech, often exhibit self-monitoring behavior, attempting self-correction and expressing frustration with their grammatical errors (Oomen et al., 2005).

Unstable or missing form–value relations, detected during processing, can trigger this response. The response is a speaker-level heuristic for detection, not the definition of grammaticality. That separation explains why marginal constructions elicit gradient certainty, why repeated exposure can attenuate negative responses without necessarily changing status, and why conventionally licensed constructions can still feel wrong under processing pressure. In that sense, (un)grammaticality can be illusory (Fanselow, 2021).

Distinguishing grammatical status from subjective ratings. Grammaticality research has to distinguish grammatical status, the conventional, population-level property made precise in §3.3, from the subjective read-outs that speakers provide. Those read-outs have two channels in the formal model: anomaly and confidence. (I use "objective" below only in that sense, population-level and convention-constituted, not judge-independent.)

Table 1 summarizes the two levels of analysis.

Table 1: Grammatical status versus subjective read-out

Level	Nature	Observable through
Grammatical status	Population licensed-assembly state, estimated for a specified situation	Converging evidence: corpora, production, repair behaviour
Subjective read-out	Anomaly and confidence	Ratings, online measures, self-reports, test–retest, explicit confidence

The measurement problem is familiar from other sciences: temperature can't be directly observed but has to be inferred through the behaviour of thermometers. Grammaticality likewise has to be inferred from several measures, with acceptability ratings as one imperfect window. The same caution applies to language-model probabilities: raw string probability by itself doesn't separate grammatical from ungrammatical strings, and minimal-pair probability contrasts are informative only when the intended message is controlled (Hu et al., 2026).[3]

[3]Recent real-patterns work on language models raises the broader question of where linguistic patterns reside–in outputs, cognition, model-internal compression, or a plurality of models (Nefdt & Ladyman, 2026). OVMG doesn't settle that question for language as a whole. Its narrower claim is that grammaticality, for the purposes tracked here, is a projectible pattern in situation-indexed licensing profiles.

This status/read-out distinction also changes how factor analyses in experimental syntax should be interpreted (§3.7). Such analyses target the structure of the subjective read-out, not grammatical status directly. A factor counts as evidence about the grammar only when it also appears in the independent indicators for licensing, compatibility, and confidence; the supplement states the corresponding observation model.

This also explains why satiation and gradient judgments don't automatically threaten categorical status. A construction can remain unlicensed in the community's system while triggering progressively weaker negative responses through familiarization. Conversely, subjective response can remain continuous even when the population licensing state is effectively categorical (§4.3). Treating ratings as evidence rather than definition keeps grammatical hypotheses empirically vulnerable without collapsing the linguistic system of a communicative situation into individual responses to it.

Misattribution effects. Sometimes the subjective read-out of ungrammaticality doesn't accurately reflect conventional grammatical status. Both false positives (grammatical constructions feeling ungrammatical) and false negatives (ungrammatical constructions escaping detection) occur systematically.

When grammatical constructions feel ungrammatical. Listeners and readers can misanalyze utterances, triggering feelings of ungrammaticality even though the construction conforms to standard patterns:

(20) *The old man the boats.* (Ritchie & Thompson, 1984)

On first reading, one might treat *old man* as a noun phrase and arrive at nonsense. But the sentence has *the old* as the subject (meaning 'the elderly'), and *man* as a verb. Interpreted correctly, the sentence means 'The elderly operate the boats', and is fully grammatical.

Processing overload provides another source of misattribution. Heavily nested structures like (15) are grammatically well-formed but trigger negative responses due to cognitive limitations rather than grammatical violations.

When ungrammatical constructions escape detection. Conversely, conventionally ungrammatical utterances may fail to trigger negative responses when the intended meaning is strong:

(21) * *In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The intended interpretation is clear, and hearers readily infer the meaningful comparison. Under the relevant syntactic analysis, the comparison composes differently. The first clause suggests comparing numbers of people, but the second clause has *they* ('more people'), making the meaning 'more people did x than more people did y', which is nonsensical. But this violation often escapes detection.

Agreement errors can similarly hide within complex noun phrases:

(22) * *The patchwork of laws governing background checks, addressing assault-weapons limits, and regulating open-carry practices help explain why people continue to be wounded and killed.* (modified from Philip B. Corbett's editing quiz Corbett, 2016)

The singular subject *patchwork* clashes with plural *help*, but the intervening plural nouns mask this violation. Cognitive resources devoted to processing complexity can prevent speakers from registering the agreement error.

2.5 Mechanisms of grammatical change

The stability of form–value relations isn't static. Various motivations change which forms speakers choose and which new or established forms they retain, explaining how grammatical systems evolve over time. These aren't a loose catalogue of causes: each names a route by which learning or population transmission can change, so each predicts which forms should gain licensing and which should lose it.

Semantic motivations for reanalysis. Speakers regularly deploy metaphors, analogies, and context-induced reinterpretations that stretch or shift the meaning of existing forms. Over time, such re-analyses can yield new grammatical constructions that weren't previously part of the language's repertoire.

A well-documented example is the development of the *going to* futurate construction. Historically, *going to* described physical motion toward a place:

(23) *I am going to London.*

Frequent usage in contexts where motion implied subsequent action (e.g. *I am going to fetch some firewood*) led to semantic shift. The directional component was reinterpreted as marking future intention rather than spatial goals:

(24) *It's going to rain.*

What once expressed physical trajectory now functions as a predictive futurate marker. The form–value relation changed through semantic reanalysis, driven by communicative utility.

Social motivations for licensing and excluding forms. Linguistic variants often serve as markers of regional background, class, ethnic identity, or age group. These social cues motivate speakers to favor some forms over others; over time, those preferences shift what counts as grammatical.

The decline of the simple past (*le passé simple*) in modern spoken French illustrates social motivation. Forms like *il alla* ('he went') became associated with formal, literary, or provincial speech. To avoid signaling undesirable social affiliations, speakers gravitated toward compound forms like *il est allé*, which lacked these status-laden connotations.

Situational licensing tracks changing social dynamics: forms that were once fully grammatical can become socially marked and gradually excluded from normative grammar.

Structural motivations. Structural considerations arise from cognitive and communicative demands, including processing limitations, the need for clarity, and the influence of analogy.

Processing constraints and memorability. Transmission tends to favor patterns that minimize processing demands. Forms requiring multiple long-distance dependencies or heavy embedding are less likely to achieve stable transmission across generations. This pressure toward processability shapes grammatical evolution, with simpler patterns spreading through communities.

Analogical extension. When speakers encounter new communicative challenges, they often extend known patterns to new contexts:

(25) ? *I asked about what did she do.*

Such forms are licensed in some English varieties and marginal or rejected in others. Where they spread, the analogical source is readily identifiable: main-clause interrogatives such as *What did she* *do?*. If licensed in a new community, this analogical extension would represent structural motivation reshaping the grammar.

Iconic motivations. Forms are sometimes licensed precisely because their structure reflects their meaning. Reduplication for intensity provides a simple example:

(26) *a big big problem*

The formal repetition iconically mirrors the magnitude being expressed. Such patterns are readily interpretable and likely to spread because the form–value link is immediately apparent.

Iconicity remains syntactically and variety constrained, though. In the target standard-English frame here, adjectival reduplication is natural in pre-head modifiers but rejected as a predicative complement:

(27) * *the problem was big big*

These motivations (semantic, social, structural, and iconic) can interact and sometimes conflict. While Optimality Theory has productively modeled such competing pressures through ranked constraints (Prince & Smolensky, 2004), OVMG views their resolution as arising from usage within communicative situations rather than solely from fixed rankings. The specific weighting of motivations reflects historically established patterns and communicative needs.

3 From an utterance to a grammaticality profile

The preceding section supplied diagnostic labels. This section states the model they belong to without asking the reader to learn its mathematical representation. The full specification, derivations, simulations, and notation are in the formal supplement. The main idea is simple: a grammaticality claim compresses several answers about a form–value pairing in a particular community and situation. The answers can come apart, and their pattern matters more than any single score.

The model separates a state theory from a dynamics theory. The state theory asks what has to be true for a candidate to count as grammatical now. The dynamics theory asks how learning, choice, competition, repair, and forgetting could make that state persist or change. This separation blocks a common shortcut: a story about how a pattern arose doesn't by itself establish the pattern's present status, and a stable present pattern doesn't identify the mechanism that produced it.

3.1 Fix the situation before diagnosing the form

Every diagnosis is relative to a conditioning state: the communicative situation as the participants construe it, together with the community, variety, register, or norm centre to which they orient. This state can include the activity, medium, footing, attributed speaker identity, and relevant discourse community. It isn't assumed to be obvious or externally given. Interlocutors can construe the same encounter differently, and analysts can draw the boundary too broadly or too narrowly.

For that reason, the conditioning state has to be specified independently of the judgment the model is meant to explain. A study can't label every accepting speaker as belonging to one variety and every rejecting speaker as belonging to another after seeing the responses. It needs prior evidence from community membership, usage, interactional cues, self-identification, or a registered coding protocol. If an independently motivated partition turns an apparently divided population into internally stable groups, the original conditioning state was underspecified. If no such partition does, the disagreement is a genuine population fact.

3.2 Four constitutive questions

OVMG treats constructions in the broad Construction Grammar sense: learned, typed associations among form, structure, content, context, and conventional constraints, at any grain from an inflectional cell to a clause pattern or an intonational contour (Fillmore et al., 1988; Goldberg, 1995; Sag, 2012). A candidate utterance can normally be assembled in more than one way from such resources. The model asks whether *some* analysis of the intended form and value passes four tests.

1. Coverage: can the language build it? There has to be at least one complete analysis of the form with the intended value. Word salad may fail here. A merely novel sentence need not: productive constructions can cover a token that has never previously occurred.
2. Compatibility: do its operator instructions fit together? The contributions that tell interlocutors how to update the conversation have to be jointly satisfiable. In **I've finished it yesterday*, the present perfect and the deictic past-time adverb provide incompatible temporal instructions. More exposure to the same analysis can't average that conflict away.
3. Saturation: are the contrasts required in this niche supplied? A community can come to treat overt marking of a distinction as obligatory. On this account, obligatoriness isn't simply listed in advance. It's the historical result of situations in which an unmarked alternative repeatedly loses to an overtly marked one. A moribund distinction can later cease to be obligatory.
4. Licensing: are the constructions used in the analysis established resources here? A form can be analyzable, interpretable, and internally compatible without being part of the relevant repertoire. English **I have 25 years* illustrates a head-and-complement convention of this kind. The intended age meaning is recoverable, but English licenses a different frame. No operator has to be mis-set for a construction to fail this test.

These tests are ordered for diagnosis but aren't four degrees on one scale. Coverage and compatibility are properties of analyses in a specified inventory. Saturation records a path-dependent community convention. Licensing is a population pattern over constructional resources. A candidate is grammatical when at least one analysis passes all four; alternative failed analyses don't contaminate a successful one. This scope matters for ambiguity, coercion, and productive novelty.

The distinction between operator value and exponent choice cuts across the four tests. **Goed* expresses the intended past value but selects an unlicensed exponent. Turkish suffix-harmony errors can have the same shape: the plural or past value remains identifiable, while the chosen allomorph isn't licensed in that phonological environment. Conversely, **depend of* is a head-specific complement convention, not an operator error. Such cases show why neither closed-class status nor categorical rejection defines the operator stratum.

3.3 Population status and a speaker's evidence

The four questions concern a population profile, but no speaker or analyst sees that profile directly. Each observes a history of tokens, alternatives, omissions, repairs, and contextual cues and forms an estimate from it. OVMG keeps three things distinct: the community's current licensing pattern, an individual speaker's evidence-based estimate of that pattern, and an analyst's estimate from a possibly different sample.

This distinction adds an important second dimension to the familiar gradient mean. Two forms can receive the same middling rating for quite different reasons. One speaker may have little evidence either way; another community may be sharply divided between stable users and stable non-users. The first is epistemic uncertainty and should be sensitive to framing and new exposure. The second is population heterogeneity and should instead appear as stable between-speaker disagreement. Mean ratings alone erase the difference.

The model also distinguishes uncertainty about the constructional inventory from uncertainty about licensing. If analysts don't know whether a productive analysis exists, that's an uncertainty about coverage. Once a viable analysis is fixed, sparse evidence about whether a community uses its nodes is a licensing question. Treating every unfamiliar token as absent from the inventory would wrongly predict that first-heard but readily productive sentences begin as ungrammatical.

3.4 What the profile is expected to predict

The profile earns its theoretical role only if observing part of it licenses expectations about cases not already used to define it. In plain terms, the analysis should tell us more than which label to attach to yesterday's rating. It should help predict how a form will be repaired, learned, transmitted, softened by exposure, or destabilized over time.

The central projection contrasts weak evidence with strong contrary evidence. A construction rare because its niche seldom occurs should remain uncertain and comparatively movable under informative exposure. A form absent despite many genuine opportunities, and repeatedly displaced by an established competitor, should produce confident and framing-resistant rejection. A processing-heavy but licensed sentence should improve with practice without a corresponding change in production or repair evidence. A moribund contrast should first lose evidential support and only later lose its categorical treatment. These targets go beyond the tests used to assign the initial profile and can fail independently.

Those failures have to change the analysis. The situation, niche, and target projection should be fixed before the outcome is inspected. If a profile then fails to predict held-out repair, exposure, transmission, or change, the projection has to be narrowed, split, or abandoned. It can't be rescued by redrawing the speech-community boundary after the responses are known.

The supporting explanation is deliberately weaker than a claim of self-regulation. A stable profile might be the residue of history, the result of ongoing transmission, or the effect of an active corrective process. OVMG calls these stabilization, maintenance, and control, respectively. Evidence for one doesn't establish the next. In particular, repair is only a candidate controller until intervention shows that a deviation is registered and that the response helps restore the threatened higher-level relation.

3.5 Status and the feeling of status

Speakers don't inspect a population state. They experience an utterance and respond to it. OVMG represents that response with two ordinary-language dimensions: how anomalous the token feels and how confident the speaker is in that reaction. Low expected licensing can contribute to anomaly, but so can processing load, lexical implausibility, garden paths, prescriptive pressure, and misidentification of the speaker or situation.

Confidence supplies information that anomaly alone misses. A preempted gap may elicit a confident "that's impossible". A winnerless cell may elicit an equally low rating but the phenomenology "I don't know how to say it". An independent-relative *whose* may feel unfamiliar while remaining movable under framing. Centre embedding can feel very bad even when the speaker accepts the construction once its parse is made clear. These aren't verbal glosses added after the fact: the model predicts different patterns of test–retest stability, confidence, exposure response, production, and repair.

A community or institution may impose a practical acceptance boundary on this graded evidence. That boundary belongs to a decision layer: changing the costs of false acceptance and false rejection should change how a hearer responds without changing the underlying licensing profile. A sharp switch in overt response would test the decision process, not prove that grammatical knowledge itself contains a numerical cutoff.

3.6 Misalignment and the scope of a successful analysis

Hearer and speaker can disagree about which situation is in force. A form licensed in one dialect, genre, age group, or bilingual practice may be heard through another community's expectations. The resulting anomaly can reflect a real mismatch between two stable systems rather than uncertainty within either one. Attribution matters too: a hearer who treats a token as belonging to another community, a learner error, noise, or personal overload need not respond as if it threatened the local convention.

Within a fixed situation, the existence of one successful analysis is enough. This prevents the model from rejecting ordinary ambiguity merely because one available parse fails. But the value has to be fixed independently enough to make the claim vulnerable. If analysts are free to invent a new value whenever an analysis would otherwise fail, coverage becomes vacuous. The intended value should be constrained by context, paraphrase, comprehension, and the speaker's subsequent commitments.

3.7 Measurement requires converging channels

Status isn't directly observed, so no single observation defines it. The empirical programme combines four channels:

- Production shows which available alternatives speakers choose in independently specified niches. A low corpus rate can reflect unavailability, low preference, or avoidance, so production alone can't identify licensing.
- Repair shows how interlocutors respond to a putative violation. Understanding repair, update-oriented repair, overt correction, and prescriptive sanction have to be coded separately.
- Judgment and confidence measure the subjective read-out rather than status directly. Repetition, framing, and test–retest designs show how stable that read-out is.
- Situation cues support the analyst's and hearer's identification of the relevant community and activity. They can't be inferred solely from the target rating.

A strong test requires at least one non-judgment indicator of licensing, an independent measure of whether changing an operator value changes the public update, and a niche definition fixed without using the target's acceptability. The formal supplement states the corresponding observation model and its identification assumptions. The agreement corpus probe reported there illustrates the discipline: a rate per opportunity is evidence about choice, and becomes evidence about licensing only after the alternatives and avoidance option have been independently normed.

3.8 The opening examples revisited

The profile now gives the opening asterisks different explanations. The results are summarized in Table 2; the labels describe the pivotal condition rather than mutually exclusive sentence types.

4 How licensing patterns change

The state theory says what a present profile consists of. It doesn't explain how the profile came to be. The dynamics module supplies candidate routes, with two constraints on the explanation. First,

Case	Pivotal diagnosis	Expected profile
**Can the have running*	no covering analysis	confident rejection across construals
Colorless green ideas sleep furiously	licensed but implausible content	grammatical status with content oddness
**I've finished it yesterday*	incompatible temporal instructions	confident rejection resistant to exposure of the parts
**I have 25 years*	concentrated non-licensing of the English age frame	recoverable meaning but confident, competitor-backed rejection
?friend of whose	sparse opportunity and weak evidence	low confidence and comparatively strong framing or exposure effects
**Which did you buy car?*	putative high-opportunity preemption	confident rejection only if the opportunity and competitor assumptions are independently met
pobedit' first-person singular	winnerless cell	low rating with ineffability rather than confident correction
centre embedding	implementation overload	practice can improve the experience without changing licensing

Table 2: Reader-facing OVMG profiles. Each diagnosis makes a different prediction beyond the judgment used to introduce it.

learning about a community isn't the same event as changing that community. Second, absence is informative only when the missing candidate had a genuine opportunity to occur.

4.1 Opportunity, candidates, and the outside option

A NICHE is a recurring communicative job, such as expressing a recipient, locating an event in time, or marking an information source. Its candidate set contains the realizations speakers could plausibly choose for that job at that time. It also contains an outside option: paraphrase, clarification, silence, or avoidance. The outside option matters because failure to produce a target doesn't always mean that a competitor defeated it.

Production reflects both licensing and choice. A licensed form may be rare because speakers prefer another licensed form. An unlicensed form may be absent for the same surface reason. Conversely, a form may be scarcely attested simply because the relevant niche is scarcely encountered. The analyst has to establish the opportunity set, candidate preferences, and avoidance rate without using the target's grammaticality judgment as the coding criterion.

The evidence stream distinguishes three events. A direct error provides evidence about the target itself. An omission provides evidence only to the extent that the target would plausibly have been chosen had it been available and that the niche was correctly identified. Repair provides a further observation whose meaning depends on its type and on the relation between the error and the public update. These events enter one shared model but aren't interchangeable counts.

4.2 Learning and population change

An individual updates an estimate when evidence arrives. The population changes only when that evidence alters later inclusion, production, learning, or transmission. This distinction is central. Updating an evidence-based estimate can make beliefs more precise; it can't make speakers adopt or abandon a construction. The formal dynamics add an explicit adoption-and-retention process whose costs have to be motivated independently.

That process is intentionally conditional. Under some settings, weakly supported constructions tend to disappear and strongly supported ones tend to persist, producing separated population patterns. Under other settings, the same learning rule yields a single interior pattern rather than separated outer patterns. The model doesn't derive categorical grammar from exposure alone. It identifies the extra assumptions under which categorical-looking regimes can emerge and makes those assumptions available for empirical challenge.

4.3 What the simulations establish

The simulations serve as existence and coherence checks. A deterministic approximation locates candidate long-run regimes, and a stochastic agent simulation checks whether finite populations with sampled evidence exhibit the same qualitative behaviour. Together they show that separated licensing patterns are possible when the adoption response, opportunity structure, memory, and competition are jointly strong enough. They also show that the separation disappears when those conditions are removed.

This conditional mechanism result isn't an empirical fit or a theorem that all languages polarize. The model doesn't yet establish the exact long-run behaviour of bounded-memory populations or how long they remain in one regime. Nor do the simulations currently model networked exposure, cohort replacement, or all channels through a single shared underlying state. The supplement reports these open proof and identification obligations alongside the successful checks.

4.4 Repair as a candidate maintainer or controller

Repair can feed back into later learning, but different repairs do different work. Some secure understanding without endorsing the local form. Some replace an exponent but leave the intended operator value intact. Some correct an operator mis-setting that would otherwise change who is committed to what, when, or with which discourse status. Overt correction can also reflect prescriptive or identity policing rather than a threat to the update.

The operator hypothesis predicts a selective response profile. Once closure, opportunity, competitor strength, and ordinary comprehensibility are controlled, mis-setting an update-configuring contrast should disproportionately elicit update-oriented repair. A categorically unlicensed but non-operator construction may instead be corrected as a conventional form choice. The prediction concerns the repair target. Non-operators can still be categorically policed.

Calling repair a *controller* requires more than observing that correction and stability coexist. An intervention would have to show that the system registers the relevant deviation, changes lower-level behaviour in response, and thereby restores the threatened population relation. Evidence that removing repair weakens transmission would establish maintenance; evidence of the full deviation-sensitive loop would establish control.

4.5 Four predicted trajectories

The dynamics yield four qualitatively different trajectories that would be collapsed by a single frequency measure.

Obligatorification. An overt contrast becomes obligatory when unmarked realizations repeatedly lose in well-attested niches and that pattern persists long enough to become a community convention. English progressive marking and Turkish evidential marking are proposed cases. The claim is historical and niche-specific: the model can't infer obligatoriness from the analyst's present uncertainty or from the absence of data. Once established, obligatoriness also doesn't vanish with the first contrary token; recovery has to persist before the convention changes.

Winnerless cells. A dense niche can lack a conventional winner. Speakers may paraphrase or avoid the job. Because avoidance isn't readily attributable to any one candidate, it doesn't by itself lower support for a particular candidate; it mainly starves all of them of evidence. Low licensing with low confidence also requires independently motivated weak starting support or diffuse negative evidence. The resulting profile is ineffability rather than a competitor-backed ban. Russian defective paradigms supply the clean model; English **amn't* is likely mixed because a competitor and avoidance both matter.

Destabilization. A moribund contrast can lose evidential concentration before its average status changes much. If a shared evidence stream disappears, individual histories can drift back toward starting expectations. Between-speaker divergence follows early only when a further source of heterogeneity (different networks, cohorts, starting expectations, or exposure histories) is present. Concentration loss is the direct prediction; population dispersion is conditional.

Actuation and apparent categorical exclusions. Small changes in opportunity, competitor strength, analogy, repair, or social ascription can move a construction across a boundary between long-run regimes. That supplies candidate actuation routes without pretending to predict the historical trigger from the state theory alone. At the other extreme, persistent preemption in a dense niche can generate a highly concentrated rejection even when a compatible analysis remains available. Apparent categorical exclusions don't always require an additional structural ban; they require independent evidence that the niche was rich, the target was a live candidate, and a competitor repeatedly won.

The dynamics section's empirical burden is now visible without the equations. The account succeeds only if independently measured opportunities, alternatives, repair types, evidence histories, and adoption costs predict the distinct trajectories. The formal supplement supplies the quantitative claims and reproducible simulation contract; the main paper keeps their linguistic meaning and their failure conditions in view.

5 Theoretical implications

OVMG's decomposition of grammaticality determines what the category should let us expect. If a form–value relation is grammatical in a communicative situation, it should pattern with other licensed resources in repair behaviour, transmission, satiation under exposure, and trajectories of change. If it's unlicensed, weakly licensed, or backed by little evidence, those expectations should fail in different, diagnostic ways (§3.4).

In this sense, the account is projectibility-first: grammaticality earns theoretical standing by the inferences it supports, not by the mere fact that a mechanism can be named for it (Boyd, 1999; Goodman, 1955; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). The profile proposed to support those inferences is conditioned form–value stability. The state architecture yields projective tests and distinguishes three world-side support hypotheses. The full map represents conditional sufficient conditions for stabilization. The production–repair loop is a candidate maintainer whose removal effect needs longitudinal or intervention evidence. Repair is also a candidate controller only if it registers departures from the licensed profile and pushes the community back toward it. That stronger claim depends on the intervention signature in §4.4 (Reynolds, 2026b; Woodward, 2001, 2003).

This recasts the competence–performance relation. Licensing and feeling are coupled layers, not a competence core plus performance noise: processing and other performance factors help shape which form–value relations stabilize, and those stable relations in turn constrain performance. Their evidence streams remain separate. The identification strategy in §3.1 keeps this from becoming a catch-all, since licensing, anomaly, and confidence are measured by different evidence streams and can falsify each other. The rest of this section develops the consequences: the account's reading of macro-typological universals, its distinctive predictions, and its relation to neighbouring frameworks.

5.1 Macro-typological constraints on grammatical design

Recent macro-typological work sharpens the question of how strongly grammar is constrained across languages. Verkerk et al. (2025) test 191 implicational universals from the Universals Archive against Grambank's 2,430-language morphosyntactic sample, using Bayesian models that control explicitly for genealogical and areal non-independence and then follow up with spatiophylogenetic analyses of evolutionary rates.

A naïve analysis that ignores relatedness appears to support the vast majority of proposed universals, but once phylogeny and geography are accounted for, only about a third (60 of 191) remain statistically supported. Support is concentrated in relatively narrow domains: most hierarchical universals and a substantial minority of "narrow" word-order universals survive, whereas "broad" word-order universals and miscellaneous others fare poorly. Diachronically, supported universals typically correspond to "harmonic" combinations of features (for instance, consistent head–dependent orders) that languages are more likely to evolve into than out of, with these preferred configurations recurring independently across lineages.

From an OVMG perspective, these findings identify candidate attractors in the design space of operator-valued form–value relations rather than exceptionless grammatical laws. They don't by themselves show that a macro-typological feature is a language-internal grammatical object. The typological claim has to pass through a comparandum-to-realization mapping, which requires specifying the comparative feature is being tracked, how each language realizes the relevant operator value or form–value relation, and whether the resulting profile predicts held-out languages or diachronic trajectories (Reynolds, 2025).

Under that discipline, supported patterns correspond to configurations for which choice and independently specified adoption pressures favour inclusion across a wide range of communities. They're cognitively and communicatively favourable enough that repeated episodes of change tend to draw licensing toward the same kind of stable region in lineage after lineage.

The fact that roughly two-thirds of the tested universals fail once autocorrelation is controlled for also fits a de-idealized view of grammaticality. Community-specific conventions still have considerable freedom in how they realize operator values, with only some regions of the space strongly preferred. Large-scale comparative work of this kind complements OVMG by locating candidate stable regions; OVMG specifies conditions under which local community dynamics and operator compatibility could make those regions reachable and persistent.

5.2 Predictions

The model's distinctive predictions come from separating positive evidence, opportunity structure, how likely a form would have been chosen had it been licensed, how settled the licensing evidence is, and repair flow. Cross-linguistic gender remains a useful boundary case, but it isn't the strongest test. Any framework that treats grammaticalized concord as obligatory predicts stronger judgments where concord is more tightly grammaticalized. The companion operator-stratum account also complicates the simple gender story: gender concord can remain categorically licensed through concentrated paradigm evidence even when its public-update contribution is attenuated (Reynolds, 2026c). The sharper predictions concern concentration contrasts, zero-evidence learning, L2 transfer, moribundity, repair type, and change by re-licensing.

Satiation framing, as a rank order. Framing should matter most where the evidence is least concentrated. Once the relevant niche has been identified independently, confirmed preempted gaps and **sheeps* should show the least movement; entrenched but uncommon constructions should show more; and independent relative *whose* and clean winnerless cells such as *pobedit'* 1sg should show the most. Participants told that tokens of a low-concentration form come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told the same tokens are errors; the same manipulation should leave a confirmed preempted gap unmoved. Lu et al.'s (2024) meta-analysis of syntactic satiation in extraction from islands supplies a nearby benchmark for extraction phenomena, not a direct measurement of left-branch extraction. The dependent variables are rating change, evidence-confidence change, decision-confidence change, later repair behaviour, and elicited production.

Artificial-language learning. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target candidate is never produced. In the high-opportunity condition, competitor candidates repeatedly occupy the relevant niche and the target would otherwise be a plausible choice; in the low-opportunity condition, the niche rarely arises or the target would seldom be chosen even if available. OVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. The design operationalizes the supplement's sparse-opportunity versus dense-preemption contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 preemption transfer. Early L2 rejections should track L1 preemption mass, not just L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, their situation ascriptions and constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker's (1972) concept of interlanguage, but gives a more specific prediction: transfer is strongest where the L1 has accumulated many attributable, counterfactually informative omissions.

Concentration loss before categorical decline. For a contrast going moribund, the direct destabilization result (§4.5) is loss of excess evidence concentration as opportunities thin. Longitudinal designs should measure confidence, test–retest stability, and sensitivity to controlled exposure alongside means and production. A reliable drop in those concentration-sensitive measures while the population rate remains approximately stable would support the proposed precursor. Inter-speaker judgment dispersion is a separate, conditional target: prior heterogeneity by itself makes normalized dispersion lag the mean, so a dispersion-leading result should be predicted only when cohort replacement, unequal network opportunity loss, heterogeneous retention, or a specified judgment mapping supplies faster divergence.

Soft alternations as a non-bimodal control. The conditional dynamics (§4.3) predict separated licensing only where the observation process and independently motivated adoption costs support it; a both-licensed alternation should stay unimodal, its variation carried entirely by selection among two licensed variants and any outside option. The dative is the worked case (§5.5): a corpus production model is well calibrated on the recipient choice within its high-frequency core, the graded, single-peaked signature OVMG predicts when both variants are securely licensed. The alternation is non-categorical through optionality, not the uncertainty of a winnerless cell (§4.5): the niche is common and both variants are licensed, so judgments should be confident and gradient rather than ineffable. An account that reads categoricality off structural membership predicts the opposite.

Repair type. The repair streams (§4.4) predict which part of an utterance is targeted, not whether every non-configuring form is merely stylistic. Update-oriented repair should increase with niche opportunity and with the extent to which changing the contrast changes the public update, conditional on the other factors. Form correction can remain strong for a categorically excluded construction or exponent even when the intended public update is preserved. Opportunity, update consequences, and licensing concentration are independently estimable from opportunity-annotated corpora, comprehension probes fixed before the repair data is inspected, and the non-rating licensing streams. If matched non-configuring contrasts draw the same update-oriented repair as operator contrasts, or if that repair tracks social indexing rather than measured update consequences, the operator-directed controller claim fails.

Change by re-licensing. Hard unification makes a commitment that a weighted-penalty model doesn't: no amount of entrenchment can make the same clashing operator assignment grammatical. Apparent change in cases like **I've finished it yesterday* should come either from read-out effects or from re-licensing. Where present-perfect combinations with definite past-time adverbials are attested in a community (Werner, 2013a, 2013b), change has to proceed by *re-licensing*: a perfect construction with a different operator contribution enters the repertoire, potentially producing judgment distributions that are bimodal across communities and categorical within speakers, not gradual within-speaker reduction in the status cost of the original clash. A soft-constraint model predicts the opposite signature. Judgment-distribution data from the relevant contact varieties decides it.

This contrasts with preempted gaps. A preempted gap can re-open when changed preferences, opportunities, or target evidence move the population across a transition boundary: average licensing then rises through the middle, producing within-speaker gradience during the transition and potentially a community S-curve.

Because production supplies later evidence and retained evidence affects later adoption, some settings permit a stronger, history-dependent pattern. Two communities under the same current external conditions can remain stably different because their inherited evidence states place them on different trajectories. Licensing can then disappear at one point as conditions worsen but return only after conditions improve beyond a different point. This history dependence predicts distinct loss and recovery boundaries.

The prediction is conditional and requires longitudinal evidence. Effective exposure and the evidence speakers require before adopting or retaining the form have to be estimated without using the transition points; if matched loss and recovery trajectories retrace the same boundary, a steep but reversible response is the better account.

Compatibility failures can't re-open that way; they require a distinct construction with a different operator contribution. The identification discipline is the same as elsewhere: the re-licensed construction has to be diagnosed before the judgment data is used, for example by showing that it supports other definite past-time adverbials, scope interactions, or ellipsis patterns predicted by the new operator contribution.

Intonational operators. Nothing in the model privileges segmental material, so intonational patterns are predicted to count as operator exponents wherever they form closed, update-configuring repertoires. Opportunity density affects the concentration of their licensing evidence, not membership. The English polar-question rise and nuclear-accent placement configure what an utterance does to the common ground much as tense and agreement do (Reynolds, 2026c). The Turkish suffix-harmony analysis in Appendix A gives the parallel case for segmental exponent choice: phonological material matters for grammaticality when it realizes an operator exponent. The corresponding intonational test compares repair targets. After opportunity and competitor strength are matched, mis-set intonational operator values should produce more update confusion and update-oriented repair than value-intact exponent errors or non-operator accent shifts. Categorical correction and resistance to framing can occur in any of those cases, so neither response diagnoses operator membership by itself.

5.3 Relationship to generative grammar

Generative work made a central contribution by showing that grammaticality can't be reduced to semantic plausibility, processing ease, or frequency of attestation. Chomsky's *Colorless green ideas sleep furiously* made the contrast visible: speakers can recognize syntactically well-formed sentences even when they're semantically bizarre. Generative analyses explain apparent categorical exclusions like the determiner–head discontinuity in English left-branch extraction (**Which did you buy car?*), and they capture the fact that such sequences remain unacceptable even when their intended meaning is clear and processing demands are low.

OVMG retains the lesson. Judgments reflect systematic patterns (Dennett, 1991); those patterns can't be reduced to meaning or processing alone; and certain syntactic configurations appear to be excluded regardless of context. The earlier diagnostics for center embeddings and preempted gaps build on ideas about recursive structure and acknowledge the generative discovery that some exclusions are systematic. They relocate the explanation in coverage, hard compatibility, saturation, and concentrated exclusion.

The same point governs the account's use of formal evidence. A generative analysis can supply a real warrant when it specifies the target, diagnostics, evidence, and failure conditions for a grammatical contrast. OVMG isn't an attempt to replace every I-language analysis; it rejects only the stronger claim that categorical grammaticality has to be explained by an autonomous syntactic component or a UG-backed inventory of primitives. Its target is the community-level licensing profile that makes some form–value relations categorically licensed or excluded.

One well-known case is independent relative *whose* (1d). As Hankamer and Postal (1973) observe, the construction appears to violate no syntactic principles: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and *whose* can be used as a dependent relative pronoun (*the student whose friend was visiting*). The generative tradition's careful documentation of such cases, where seemingly parallel constructions show puzzlingly different grammatical status, has driven theoretical development.

OVMG explains the contrast differently. Rather than positing an autonomous syntactic component, it treats grammatical constraints as products of form–value relations within specific communicative situations. For independent relative *whose* to be felicitous, multiple conditions have to converge: the possessor has to be sufficiently accessible in the discourse while the possessum is predictable enough to license ellipsis, but the possessive relationship needs to be semantically significant enough to warrant explicit marking, and this configuration has to occur in a context where a relative clause is the natural way to package this information.

The relevant niche is tiny. Speakers encounter the construction so rarely, despite perfectly common components, that the licensing evidence remains thin; even when all conditions align, the construction feels alien because the evidence is thin, not because the community has accumulated confident negative evidence against it.

The difference is evidential rather than merely terminological. A generative theory has to explain why a syntactically possible and pragmatically useful construction remains marginal. OVMG predicts the profile of that marginality: low concentration, framing sensitivity, and weak confidence, not the non-satiating rejection profile of a preempted gap. The result preserves the generative recognition of systematic constraints while embedding it in a theory of how form–value relations become established and maintained in language communities.

Other cases that generative grammar struggles to explain are grammatical with one meaning but ungrammatical with another, such as *have*+numeral years. While syntactically identical to grammatical expressions like *I have 25 dollars*, this construction becomes ungrammatical specifically when used to express age. A purely syntactic account would have to explain why the same structure is well-formed in one case but ill-formed in another, despite no apparent syntactic differences.

OVMG, in contrast, locates the source of ungrammaticality in the community's form–value conventions: *have*+numeral years is licensed for duration or future time (*I have 16 years until retirement*), but the age-predication use is driven toward zero in English by dense competition with the *be*+age frame (*I am 16 years old*). Similar cases arise with plural forms that are grammatical with some meanings but not others (e.g. *peoples* for ethnic groups but not multiple individuals) and with verbs that resist certain arguments despite no obvious syntactic prohibition (e.g. *discuss about*). These meaning-dependent grammaticality patterns suggest that licensing is keyed to specific form–value associations rather than purely structural templates.

5.4 Relationship to Construction Grammar

Construction Grammar (CxG) (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012) made a central contribution by treating learned form–meaning pairings as grammatical objects. At its core, CxG argues that language consists of learned relationships between form and meaning at multiple levels of complexity. These form–value relations, or constructions, include both individual morphemes and abstract syntactic patterns. This perspective helps explain phenomena that proved challenging for earlier approaches, including idiomatic and partially schematic expressions that don't fit a clean core/periphery split.

CxG shows that meaning suffuses all levels of grammatical organization. Rather than treating syntax as an autonomous formal system that interfaces with semantics only at designated points, CxG reveals how meaning and form are inseparable aspects of linguistic knowledge. For instance, the *What's X doing Y?* construction carries an implication of incongruity that can't be derived from its component parts (Kay & Fillmore, 1999). Recent work connects this framework to model-internal analysis: Weissweiler et al. (2023) use construction grammar to probe how neural language models handle different levels of linguistic abstraction.

OVMG shares these CxG commitments about form–value relations and constructional abstraction. It doesn't privilege one expressive substance over another. The companion operator-stratum paper explicitly denies a substance-based boundary: phonological, gestural, lexical, morphological, and syntactic material can all realize operator contrasts when they enter a closed, accountable public-update repertoire (Reynolds, 2026c). The OVMG claim is narrower. Operator membership belongs to closed paradigms that configure update, role allocation, scope, or reference. Categorical licensing is broader: dense competitor evidence can also exclude a non-operator construction.

The contrast becomes clear when speakers judge violations. Register, politeness, genre, and many lexical choices can be stable and socially important without blocking the public update itself. A wrong tense, agreement value, or clause-linking marker can mis-set the update instruction; a wrong obligatory allomorph can instead fail as an unlicensed exponent choice for a closed operator. The age-predication example (*I have 25 years*) has a readily recoverable intended meaning but isn't licensed for that English frame; its rejection comes from non-licensing in a constructional niche with many opportunities, not from the substance of syntax or morphology as such.

OVMG preserves CxG's systematic treatment of constructions while separating constructional licensing from operator role. Closure, opportunity structure, competitor evidence, and independently motivated adoption costs determine whether a licensing state can occupy a separated regime (§4.3). Public-update divergence predicts a narrower comprehension and repair profile within that broader space. A construction can be categorically unlicensed without being an operator, and an operator contrast can be low-opportunity and unstable without losing its functional classification.

5.5 Relationship to usage-based approaches

OVMG shares with usage-based approaches (UBA) (Bybee, 2006, 2007, 2010) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context.

Several pieces of the model come directly from this tradition. The speaker/population split restates Schmid's (2020) separation of ENTRENCHMENT, the cognitive reorganization of a speaker's knowledge, from CONVENTIONALIZATION, the social establishment of a community's regularities. A speaker's own repertoire, the population licensing rate, and an analyst's estimate of that rate are separate objects. The proposed population change process is motivated by Croft's (2000) utterance-selection theory, in which variants are replicated and selected within a community; characteristic trajectories of change are already worked out in that lineage (Blythe & Croft, 2012).

Frequency per opportunity is likewise not new: it's the core of Stefanowitsch's (2006) case that corpora contain negative evidence, where a form counts as "significantly absent" rather than "accidentally absent" only once its observed frequency is weighed against the frequency expected given its opportunities. Variationist sociolinguistics has normalized by opportunity since Labov's principle of accountability (Labov, 1972).

OVMG adds their joint formalization and a prediction. It separates a candidate's availability from a speaker's choice among available candidates. That division puts a quantity on the absence that Stefanowitsch left open: he showed how to establish that a form is significantly absent but stressed that significant absence "does not, in itself, provide any clues as to why". A significantly absent form may be unlicensed, licensed but dispreferred, or avoided together with its competitors, and those diagnoses project differently. The formal supplement states the factorization and the restricted conditions under which an attributable omission provides approximately direct evidence of non-licensing.

Speaker-level evidence estimates feed the adoption–retention process described in §4.3. The relation to Goldberg's (2019) COVERAGE and statistical preemption is one of formalization in the same spirit. Goldberg's "coverage" is overall frequency support; it isn't this paper's structural coverage, which asks whether a complete analysis exists. Her "explain me this" puzzle (a high-coverage double-object frame preempted by the prepositional dative) is a preemption-mass story in which the competitor wins a high-opportunity niche.

The dative alternation gives the licensing-versus-selection factorization its cleanest empirical case, because both variants are licensed and the whole contrast lives in selection. Bresnan et al. (2007) model the choice between *give Kim the book* and *give the book to Kim*; a frozen version of such a model can be rescored on later British conversation in the Spoken BNC2014 to estimate selection directly. Because both realizations are grammatical, nothing in the corpus probability bears on grammatical status: it estimates the probability of an NP recipient conditional on a token that has already entered the annotated alternation sample, a conditional production probability rather than a probability of grammaticality. The released rows fix realization inside that annotated frame, the innermost of a set of nested opportunity sets; the broader licensing set is a different denominator, and reaching it marks a new target rather than a larger sample of the same one.

The conditioning that transports is projectible in Goodman's sense (Goodman, 1955): length, animacy, definiteness, and pronominality carry across varieties, while verb base rates travel worse, and even the transportable relations become overconfident out of domain. A relation can be projectible and still project poorly under a population shift, because selection is variety-relative.

The both-licensed case complements the preempted double-object gap above: on the same construction, the account separates a high-opportunity cell where an entrenched competitor repeatedly wins from a fully optional contrast where selection is the whole story, and a corpus production model measures the second without licensing a reading of the first. What a production model can't do is diagnose such a cell on its own: from inside the annotated frame, an unlicensed gap and a merely dispreferred one look alike, and telling them apart needs evidence of another kind, not more tokens of the same sampling design.

One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single competition process. OVMG keeps them analytically separate as direct error observations and counterfactual choice observations. They have to be analysed jointly rather than added as interchangeable failure counts (§4.2). Whether one underlying competition process accounts for both remains an empirical question rather than a stipulation.

The analysis of independent relative *whose*, as in ?*I saw Joan, a friend of whose was visiting*, is a case in point. A simple UBA account might predict that this construction's marginality stems from its low frequency. But this explanation proves insufficient: the construction is rare, and it's dramatically rarer than one would expect given the frequency of its component parts. Independent *whose* appears in interrogatives (*Whose is that?*), and the relative *whose* is common in dependent contexts (*the student whose paper was late*). Given these frequencies, analogical extension should make the independent relative more common than observed.

Raw component frequency is the wrong predictor; *whose* isn't a preempted gap. The frequency of independent and relative *whose* taken separately doesn't create opportunities for the niche that requires both at once, together with an accessible possessor, a predictable possessum, and a licensing ellipsis site; that convergence is vanishingly rare. This isn't the significant absence of a preempted form, which needs many opportunities and a competitor that keeps winning; it's the low-opportunity uncertainty described in the diagnostic categories above. The tension between "rarer than expected" and "uncertain rather than rejected" dissolves once licensing is separated from selection: *whose* is under-*observed* because its niche rarely arises, not preempted because a rival repeatedly beats it.

OVMG doesn't exempt the operator stratum from usage dynamics; it makes those dynamics more specific. A rare form can remain grammatical when its opportunity count is tiny or when higher-grain constructions license the token by compositional inheritance. A readily interpretable candidate can become categorically rejected when the opportunity count is large, the analogical candidate would otherwise have been a plausible choice, and an entrenched competitor wins every time. The result preserves usage-based findings about frequency and entrenchment while predicting when frequency becomes preemption.

5.6 Relationship to logicality of language accounts

An alternative perspective, prominent in recent formal semantics, suggests certain types of unacceptability stem from the language faculty itself possessing a deductive system that identifies and filters sentences with logically trivial meanings (tautologies or contradictions) (cf. Chierchia, 2013; Del Pinal, 2019; Fox, 2000). This "logicality of language" hypothesis attempts to explain, for example, systematic restrictions on quantifiers by arguing the unacceptable cases are "L-trivial", meaning their triviality arises solely from the meaning and configuration of logical or closed-class terms (like *every*, *some*, *not*), irrespective of the open-class words (like *student*, *run*) (Gajewski, 2009).

A key challenge is explaining why simple tautologies or contradictions (e.g. *It's raining and it isn't raining*) are often acceptable. Del Pinal (2019) argues against "Logical Skeletons" (which assume the system ignores open-class word identity) in favour of "LF+RESCALE", a view where the system sees standard logical forms but allows optional, context-dependent modulation of open-class terms (e.g. interpreting the second "raining" as "raining hard") to yield non-trivial meanings.

OVMG offers a broader account of ungrammaticality. "Logicality" approaches explain restrictions tied to logical and closed-class vocabulary through L-triviality. OVMG also covers cases less easily reducible to logical contradiction, such as absent form–value relations (1a), strong deviations from conventional community forms (17), and extreme, unexpected rarity (1d). By grounding grammaticality in community-specific conventions while separating hard compatibility from evidence-based licensing estimates, OVMG accommodates cross-linguistic variation and rating gradience, which are less central to the L-triviality filter.

LF+RESCALE provides a specific mechanism for acceptable "trivialities". OVMG treats those cases as possible products of more general interpretive principles within community norms, without positing a dedicated deductive module or specific operators like RESCALE. It also keeps conventional grammatical status separate from subjective processing effects or "feelings" (§3.5). OVMG

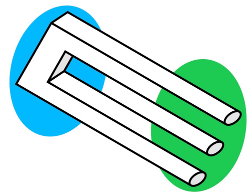

Figure 2: Impossible trident

frames grammaticality as an emergent consequence of communicative practice in the formal sense of §4.3, rather than a direct output of logical computation within syntax.

5.7 Relationship to relevance-theoretic accounts

Recent work by Scott-Phillips (2026) makes a useful observation: linguistic intuitions about acceptability arise as byproduct effects of cognitive systems for interpreting communicative acts. Just as people immediately sense when a visual stimulus violates core assumptions about physical objects (as with impossible objects), they detect when utterances violate basic presumptions about communicative efficiency. Scott-Phillips argues that unacceptability occurs not from mere inefficiency, but from an inherent impossibility of interpreting an utterance consistently with these presumptions, much as an impossible trident (Figure 2) can't be interpreted as physically cohesive in any context.

OVMG shares several premises with this account. Both reject the need for an innate grammar faculty, locating linguistic intuitions instead within general cognitive systems, and both treat language as developing from communicative needs rather than autonomous syntactic principles. OVMG's emphasis on situation-specific form–value relations builds directly on Scott-Phillips's arguments about how communicative pressures shape linguistic conventions.

The frameworks differ primarily in their explanatory mechanisms. Where Scott-Phillips argues that grammaticality judgments reduce to impossibilities of efficient interpretation, OVMG suggests that while communicative pressures shape which form–value relations become conventionalized, these relationships then create systematic constraints that can't be reduced to efficiency alone. For Scott-Phillips, one would have to demonstrate that (1c) contains inherent contradictions making efficient interpretation impossible. OVMG instead analyzes how *yesterday*'s deictic temporal anchoring fails to unify with the present perfect's reference-interval contribution. While these analyses might ultimately converge, it remains unclear why the criterion of inherent impossibility of interpretation should apply specifically to operator-value violations rather than to lexical-lexical conflicts or certain phonological patterns. The scope of what constitutes an interpretive impossibility requires further theoretical development.

Empirically, OVMG requires tests of whether specific form–value relations are stable within a communicative situation and where operator-value conflicts arise. The challenge for relevance-theoretic accounts, as Scott-Phillips (personal communication, Dec. 16, 2024) acknowledges, lies in establishing independent, empirically vulnerable claims about what makes efficient interpretation inherently impossible rather than merely difficult. The clearest comparison would test the two accounts on novel constructions as they become acceptable or unacceptable.

6 Limitations and future directions

The account has real limits, and each points to work it invites rather than forecloses. One is its reliance on the concept of "communicative situation". Speakers move among overlapping contexts that shift with the immediate setting, durable social positions, and the norms they orient to. The account builds in this fluidity, but operationalizing these dimensions, and measuring their influence on grammatical stability, will take more precise methods. A further limit is formal rather than expository: the supplement uses an approximate evidence filter, declares rather than estimates the adoption criterion, and approximates the expected population transition. Its long-run results for bounded-memory populations remain unproven. Networked exposure, cohort replacement, and a shared observation model for production, repair, ratings, and confidence remain future work. The emergence claim is conditional on those interfaces rather than a generic consequence of learning.

A second boundary concerns stylistic variation and individual preference. Most choices that fall within grammatical acceptability aren't cases of (un)grammaticality at all. They matter for this account only when they become tied to communicative situations strongly enough to affect licensing, repair, or categorical rejection. The account still has to explain how stylistic preferences become grammatical licensing facts.

A third concerns the boundary of constructional coverage. The paper argues that many apparently arbitrary restrictions are either compatibility failures, saturation failures, or concentrated nonlicensing under historical preemption. Cases that resist those treatments should be analyzed as failures to build a well-typed covering assembly, or else as evidence that the constructional inventory has been specified too narrowly.

6.1 Methodological considerations

The development of experimental syntax since the 2000s (Schütze, 2016) has transformed how gradient acceptability is measured, letting researchers quantify subtle differences in judgments and track how they change with exposure. The framework's predictions invite corpus-based, experimental, and cross-linguistic investigation. Several avenues stand out:

1. *Corpus analysis:* Large, balanced corpora allow researchers to track frequency patterns and stability over time. Investigating rare or marginal forms (e.g. the independent relative *whose*) in corpora can reveal whether low frequency corresponds to unstable operator-value patterns or merely a sampling artifact. Comparative corpus research in multiple languages can test predictions about how community values influence grammatical distinctions.
2. *Judgments:* Controlled psycholinguistic tasks, including magnitude estimation or forced-choice paradigms, can assess rating gradience, confidence, and test–retest stability. Exposure and familiarity manipulations measure satiation effects and distinguish entrenched constraints from constructions that become more acceptable through repeated exposure. Exposure effects should be interpreted through the two-channel read-out: rating improvement without corresponding change in confidence, production, or repair behaviour is evidence for movement in the anomaly channel rather than for a change in licensing status.
3. *Processing measures:* Eye-tracking, self-paced reading, and EEG studies can identify whether some judgments arise from processing overload rather than durable grammatical constraints. If a form becomes easier to parse with practice while production and repair indicators remain stable, the effect belongs to the implementation-cost channel rather than to the community's licensing profile.

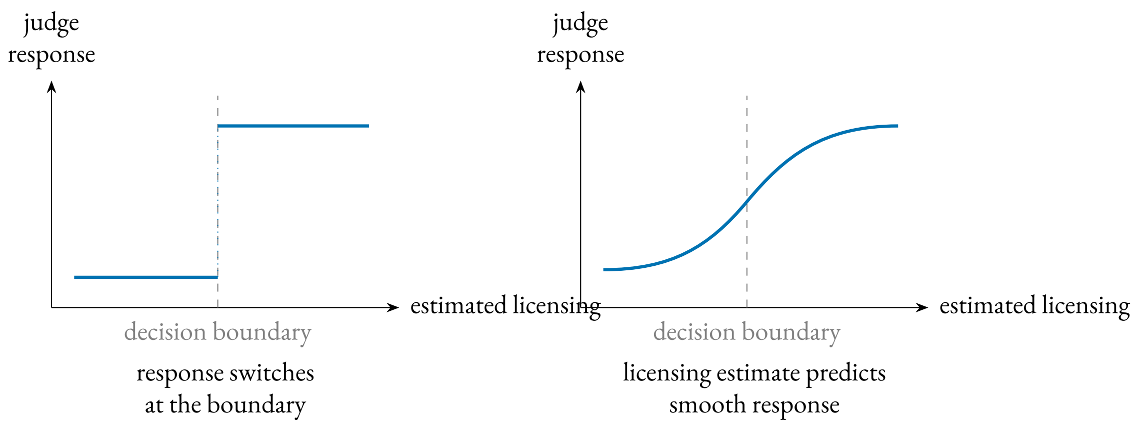

Figure 3: The decision-layer discontinuity check. If asymmetric losses induce a response switch, a judge's treatment of a form should change abruptly as the judge's estimated licensing crosses the practical boundary (left); if the read-out tracks that estimate continuously, treatment should vary smoothly through it (right). A regression-discontinuity design on multinomial hearer responses targets this decision layer, not the constitutive status state itself.

4. *Sociolinguistic fieldwork:* Investigations in communities with distinct dialects or multilingual practices can clarify how social and pragmatic motivations shape grammatical stability. Elicitation tasks and careful comparisons across speech communities can confirm that what counts as grammatical depends on local norms rather than universal principles.
5. *Longitudinal and historical studies:* Diachronic corpora and historical grammars can trace how marginal constructions evolve, testing predictions about which factors lead certain patterns to stabilize, which fade, and which undergo reanalysis. Tracking a community's licensing of previously dubious constructions over time provides direct evidence for the gradual dynamics posited by the framework.
6. *A decision-layer discontinuity check:* A community or task can place a practical decision boundary on a graded estimate of licensing. A sudden change in response type near that independently estimated boundary would test whether asymmetric losses shift hearers from acceptance to repair, request for clarification, or sanction, over and above the smooth effect of their own licensing estimate. The general design is a regression-discontinuity problem (Calonico et al., 2014; Lee & Lemieux, 2010); multinomial and covariate-adjusted versions are the needed extension (Lazzaro, 2026). In a confirmatory test, the boundary has to be estimated and registered before the outcome data is touched. A smooth response through the boundary would weaken the proposed decision-layer switch, not the licensing-state theory itself.

Figure 3 states the decision-layer check in one picture: a discontinuity is evidence for a response switch at a practical boundary, while a smooth curve is evidence that the overt response tracks estimated licensing continuously.

7 Conclusion

Grammaticality, de-idealized, isn't a categorical fact about an ideal speaker in a homogeneous community but a conditioned form–value relation. Its profile records whether some assembly covers the form, whether the assembly's operator contributions unify, whether its obligatory dimensions are saturated, and whether the population licenses its constructions. Evidence about that profile is assessed relative to a communicative situation, with uncertainty about the average pattern kept separate from genuine differences among speakers.

The competence–performance split becomes two layers with separate evidence streams: a conventional, population-level status and a two-channel subjective read-out–anomaly and confidence–identified by different measures so that they can falsify each other (§3.1). Where licensing is at issue, categoricality is conditional. Evidence updating alone doesn't create a population divide; an explicit adoption-and-retention process is also required, and only some independently motivated settings produce separated regimes. The long-run bounded-memory result remains open (§4.3). Coverage failure and hard compatibility remain distinct routes rather than default explanations for every sharp rejection.

What the profile is for is projection. A category earns its standing by supporting expectations licensed by partial observation, not by the mere presence of mechanisms: repair, transmission fidelity, satiation, and trajectories of change, over the licensing state itself and how settled the evidence for it is (§3.4).

The supporting claims are tested separately. The population dynamics supply conditional sufficient conditions for stabilization; longitudinal evidence is needed to attribute an observed profile to those dynamics. The production–repair loop is a candidate maintainer, and intervention evidence would establish whether repair registers operator mis-settings and restores the licensing profile. Section 4.4 states the relevant signatures. On this account, grammaticality matters because it predicts which forms will be repaired, transmitted, softened by exposure, or resistant to change.

These commitments expose the framework to failure. The sharp tests aren't exhausted by the decision-boundary design but form a family of profile-sensitive contrasts: confidence should dissociate preempted gaps from clean winnerless cells at matched mean ratings; framing effects should be stronger with sparse, unresolved evidence than with strong evidence against the form; artificial-language learners should distinguish zero evidence from strong preemption; early second-language rejections should track first-language omission informativeness rather than second-language frequency alone; evidence for moribund contrasts should become less settled before categorical treatment declines, while dispersion-leading change requires a named heterogeneity mechanism; update-oriented and form-oriented repair should separate; and compatibility failures should change by re-licensing, not by gradual within-speaker cost reduction. Where two stable licensing regimes are independently established, loss and recovery should also occur at different external conditions rather than retracing the same transition boundary.

The account doesn't have the simplicity of a one-variable theory. It doesn't reduce grammaticality to a single scalar, a competence–performance split, or a social-pragmatic explanation. Its economy is architectural: the same recurring distinctions (coverage, compatibility, saturation, licensing, subjective read-out, opportunity structure, and diachronic support) handle cases that otherwise invite separate stories. That economy has to be earned empirically. The conditions need independent indicators, and the projections need to fail when those indicators are wrong.

It also carries a significant memory burden. Speakers need not store every utterance as a separate item, but the account asks them to retain many situation-indexed pairings among forms, operator values, constructional neighbours, opportunity estimates, competitors, and repair histories.

That inventory is large, but the framework makes it the model's data structure: the situation-indexed pairings among forms, values, neighbours, opportunity estimates, competitors, and repair histories are the summaries from which speakers estimate licensing (§§3–4). Compositional inheritance across assemblies keeps first-heard tokens from starting at zero, and future pooling across construction families, lexemes, and situations would make the burden still more estimable. If linguistic memory is presumed small in advance, stored generalizations get forced into a rule-plus-exception format, and conditioned licensing is misdescribed as residue.

The program these predictions define is measurement-first. Estimate licensing from converging non-rating indicators (corpus rate per opportunity, elicited production, repair), fix the conditioning state before predicting rather than after, and prioritize cross-linguistic contrasts whose operator status differs, since that's where the account predicts update-confusion and update-oriented repair profiles to diverge. Opportunity density helps predict where categoricality forms or dissolves, conditional on the independently motivated adoption process. De-idealizing grammaticality trades a tractable idealization for a profile that answers to evidence. It yields a category that predicts which rejections soften, which forms transmit, and which absences hold.

A Turkish vowel harmony and operator-exponent realization

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.[4] Only suffixal harmony matters for OVMG, and even there the role is indirect. Vowel backness isn't a public-update value; plural and past are. In Turkish, the exponents of those operators have to satisfy suffixal harmony.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* 'doctor' violate backness harmony, but are fully acceptable; speakers store the form, and the community licenses it consistently.

(28) doktor
doctor
'doctor' (disharmonic stem, grammatical)

Because the word contains no malformed operator exponent, a stored construction covers it, the relevant operators unify, and the word is licensed.

SUFFIXAL HARMONY: OPERATOR-EXPONENT LICENSING. Inflectional morphemes are lexicalized with an underspecified vowel; the licensed allomorph copies stem backness and, for some suffixes, rounding. Using the "wrong" vowel doesn't make the intended value unrecoverable: speakers can identify *-ler* as a plural exponent and *-du* as a past exponent. What fails is licensing of that exponent choice in a dense operator-exponent niche.

(29) a. kitap-lar
book-PL.BACK
'books' (harmonic, grammatical)

b. * kitap-ler
book-PL.FRONT
intended 'books' (suffix harmony violation)

(30) a. gül-dü
laugh-PST.FRONT.ROUND
's/he laughed' (harmonic, grammatical)

b. * gül-du
laugh-PST.BACK.ROUND
intended 's/he laughed' (suffix harmony violation)

OVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

- The plural and past constructions are securely licensed at the population level. The mismatched exponent choice isn't licensed.
- The mismatched allomorph is still identifiable as an intended exponent, so the token is covered by a candidate operator-exponent assembly. But its pivotal exponent-choice node is almost universally excluded by overwhelming preemption from harmonic suffix choices across the many plural and past opportunities. The result is concentrated non-licensing and high confidence, not empty coverage.

[4] See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arık, 2015) for experimental evidence on native judgments.

So speakers judge the word categorically wrong, not just odd-sounding, and the confidence is evidential rather than structural: every relevant suffix opportunity supports the same allomorphic choice. This keeps suffix-harmony errors apart from word salad. **Can the have running* lacks a covering assembly; **kitap-ler* has a recoverable intended value and an identifiable exponent, but the exponent choice is almost universally excluded. By contrast, *doktor* in (28) is covered by its own stored node: stem disharmony violates no operator-exponent licensing condition, so the word remains licensed and the disharmony registers only as a phonological cost in the subjective read-out (§3.5).

Localized exceptions. Some derivational suffixes (e.g. *-imsi* '-ish') are lexically marked "disharmonic". Speakers store them, and community licensing remains consistent: no exponent condition is violated, so the form remains licensed despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

Turkish suffix-harmony errors are grammaticality failures in a precise sense: they don't mis-set a public-update contrast themselves, but they choose an unlicensed exponent for a closed inflectional operator in a dense niche. Stem disharmony lowers phonological well-formedness and affects only the subjective cost vector. A counterexample to the operator-specific response claim would be a matched phonological contrast with no operator-exponent role that produced the same update-confusion and update-oriented repair profile.

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